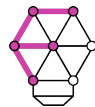


Clinical Dataset Structure: A Universal Standard for Structuring Clinical Research Datasets

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Jamie Shaffer, Julia Owen, Aaron Lee, Bhavesh Patel (presenter),
on behalf of the AI-READI Consortium



FAIR DATA
INNOVATIONS HUB





Background

Background

AI-READI project

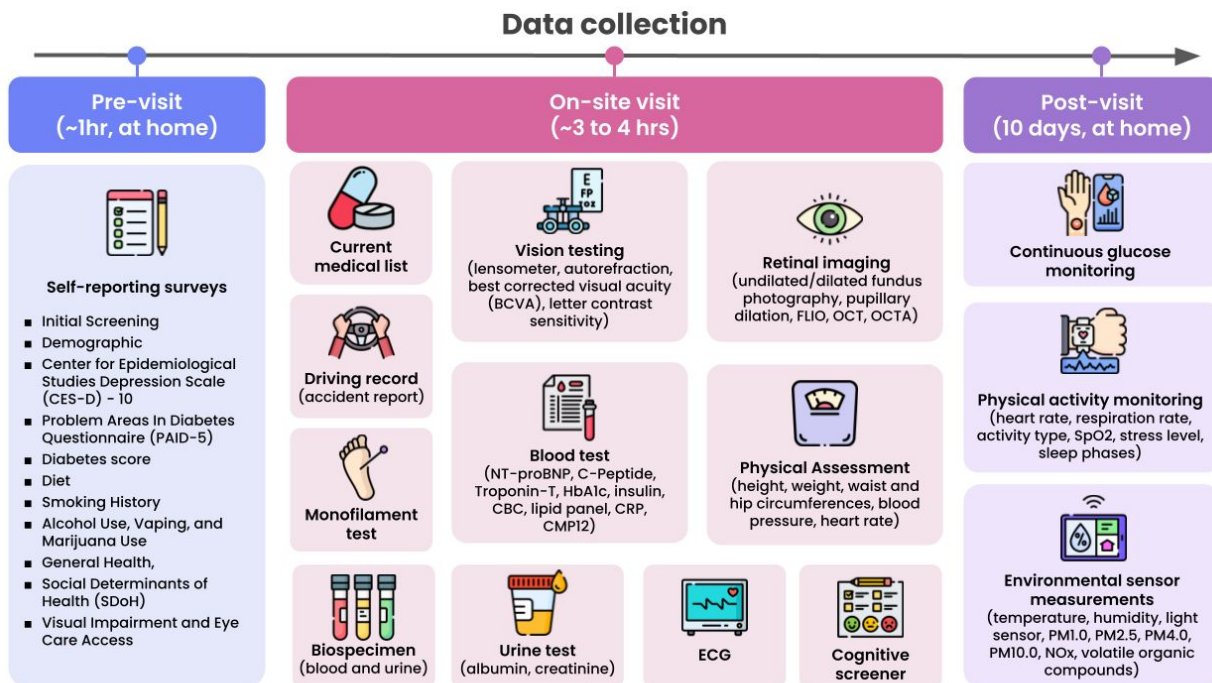
AI-READI is one of the **Data Generation Project (DGP)**
of the NIH Bridge2AI Common Fund program (2022–Present)



Goal: Collect a multimodal dataset from 4,000 participants for studying Type 2 Diabetes

Background

How to organize the multimodal AI-READI data?



100k+ data files

→ How to organize the dataset such that it is interoperable and reusable?

FLIO = Fluorescence Lifetime Imaging; OCT = Optical Coherence Tomography; OCTA = Optical Coherence Tomography Angiography, ECG = Electrocardiogram; PM1.0, 2.5, 4.0, and 10.0 = Particulate matter less than 1, 2.5, 4, and 10 microns, respectively

Background

Challenge

There is currently **no consensus on how to structure multimodal data** into a consistently organized dataset in line with the FAIR principles



Datasets are hard to reuse

Difficult for researchers other than the original creators



Datasets are not interoperable

Undermines the development of AI models



Clinical Dataset Structure

Clinical Dataset Structure

About

The Clinical Dataset Structure (CDS) is a specification to **consistently structure data and metadata** for multimodal datasets



Vision: All clinical research datasets should “look” the same

Clinical Dataset Structure

Methods

01

Review & Adapt Existing Standards

Reviewed domain-specific dataset structures to leverage existing standards.

- Brain Imaging Data Structure
- SPARC Data Structure

02

Extend for Multimodal Datasets

Incorporated missing elements using existing schemas when possible.

- Study metadata → ClinicalTrials.gov schema
- Dataset metadata: DataCite schema

03

Implement & Document

Applied specification to a real-world clinical research dataset.

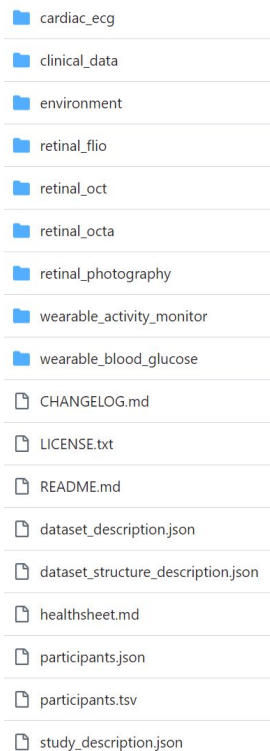
- AI-READI dataset implementation
- Detailed documentation to support reuse



Specification

Specification

Consistent folder structure



- cardiac_ecg
- clinical_data
- environment
- retinal_flio
- retinal_oct
- retinal_octa
- retinal_photography
- wearable_activity_monitor
- wearable_blood_glucose
- CHANGELOG.md
- LICENSE.txt
- README.md
- dataset_description.json
- dataset_structure_description.json
- healthsheet.md
- participants.json
- participants.tsv
- study_description.json

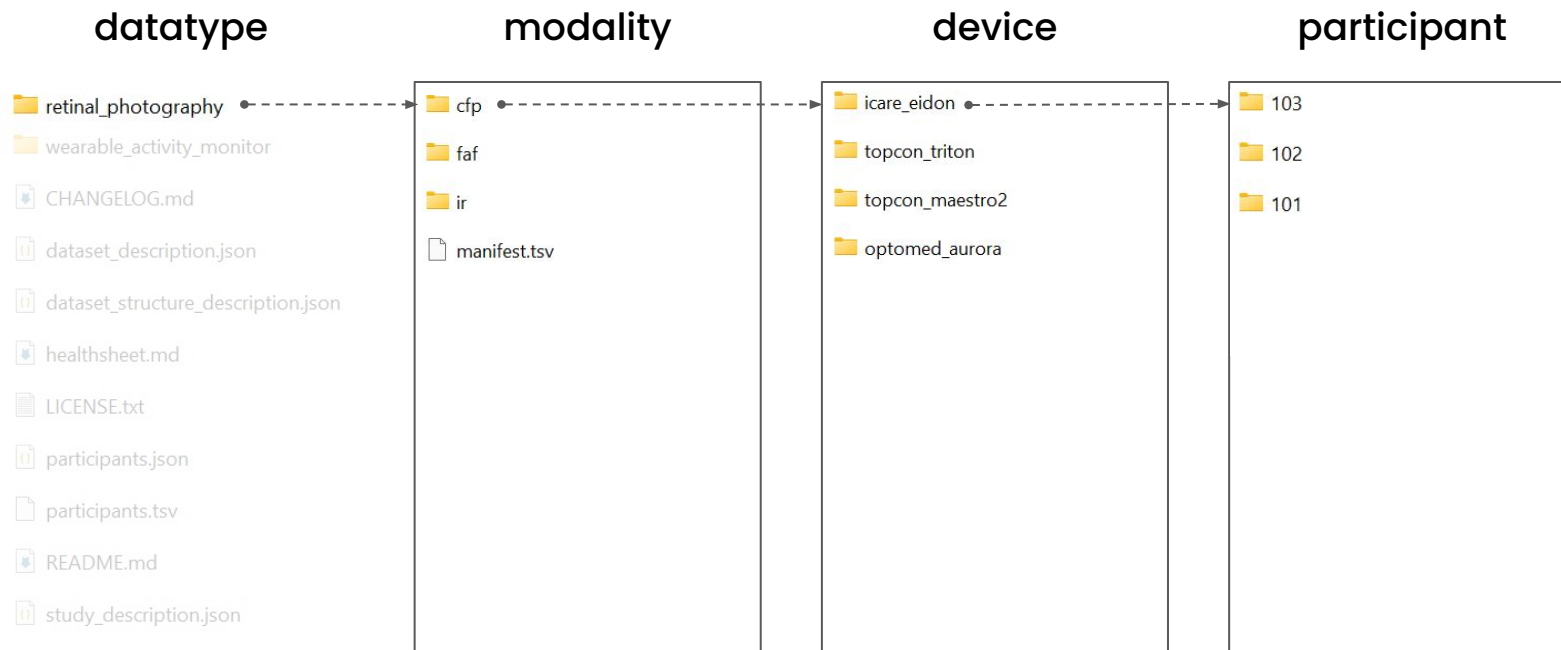
One folder per
datatype

Metadata files

Access the
AI-READI
dataset at
fairhub.io

Specification

Consistent folder structure



Specification

Standard data format

Data category in AI-READI	Data format followed
Clinical data	Observational Medical Outcomes Partnership (OMOP) Common Data Model (CDM)
Retinal imaging data	Digital Imaging and Communications in Medicine (DICOM)
ECG data (standard 12-lead)	WaveForm DataBase (WFDB)
Wearables data (physical activity monitoring, continuous glucose monitoring data)	Open mHealth
Environmental sensor data	Earth Science Data Systems (ESDS) format

Specification

Extensive metadata

retinal_photography

wearable_activity_monitor

- CHANGELOG.md
- dataset_description.json
- dataset_structure_description.json
- healthsheet.md
- LICENSE.txt
- participants.json
- participants.tsv
- README.md
- study_description.json

Broad metadata: study background, dataset provenance, dataset structure, license terms, participants information

Provided in both human and machine-friendly formats

Specification

Extensive metadata

retinal_photography

wearable_activity_monitor

CHANGELOG.md

dataset_description.json

dataset_structure_description.json →

healthsheet.md

LICENSE.txt

participants.json

participants.tsv

README.md

study_description.json

```
{
  "schema": "https://schema.aireadi.org/v0.1.1/dataset_structure_description.json",
  "directoryList": [
    {
      "directoryName": "cardiac_ecg",
      "directoryType": "dataType",
      "directoryDescription": "This directory contains electrocardiogram data collected by a 12 lead protocol",
      "relatedIdentifier": [
        {
          "relatedIdentifierValue": "https://docs.aireadi.org",
          "relatedIdentifierType": "URL",
          "relationType": "IsDocumentedBy",
          "resourceTypeGeneral": "Other",
          "relatedIdentifierDescription": "This is the documentation of the AI-READI dataset that contains add"
        }
      ],
      "relatedTerm": [
        {
          "relatedTermValue": "Electrocardiogram",
          "relatedTermIdentifier": [
            {
              "relatedTermClassificationCode": "C168186",
              "relatedTermScheme": "NCI Thesaurus (NCIT)",
              "relatedTermSchemeURI": "https://ncim.nci.nih.gov/",
              "relatedTermValueURI": "https://ncit.nci.nih.gov/ncitbrowser/pages/concept_details.jsf?dictionary="
            }
          ]
        }
      ]
    }
  ]
}
```

Specification

Extensive metadata

retinal_photography

wearable_activity_monitor

CHANGELOG.md

dataset_description.json

dataset_structure_description.json

healthsheet.md

LICENSE.txt

participants.json

participants.tsv

README.md

study_description.json

participant_id	study_group	age	study_visit_date	recommended_split	cardiac_ecg	clinical_data
101	pre_diabetes_lifestyle_controlled	51	8/22/2023	train	TRUE	TRUE
102	healthy	46	9/12/2023	train	TRUE	TRUE
103	oral_medication_and_or_non_insulin_injectable_medication_controlled	81	11/2/2023	test	TRUE	TRUE
104	oral_medication_and_or_non_insulin_injectable_medication_controlled	62	9/18/2023	val	TRUE	TRUE



Closing Remarks

Closing Remarks

Current status



The specification is fully documented

<https://cds-specification.readthedocs.io>



We are developing tools to implement CDS

<https://github.com/AI-READI/pyfairdatatools>



The AI-READI dataset is being reused

Accessed by 1,000+ researchers

Closing Remarks

Next steps



Promote the CDS

We want to evaluate interoperability with other datasets



Collect feedback and improve

Planning a survey of the AI-READI dataset users

Closing Remarks

AI-READI team



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Native BioData



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Cecilia Lee
UW



Aaron Lee
UW



Cynthia Owsley
UAB



Gerald McGwin
UAB



Shannon McWeeney
OHSU



Michelle Hribar
OHSU



Developers

Interns

Clinical Research Coordinators

Program and Project Managers

Thank You!



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fairdataihub.org

*Find these slides and
all resources here*



bit.ly/ISMB-CDS